ASSIGNMENT 1
LiDAR TOPOGRAPHY & LAZER CUTTING
ARCH 565 - Advacnced Computer Applications
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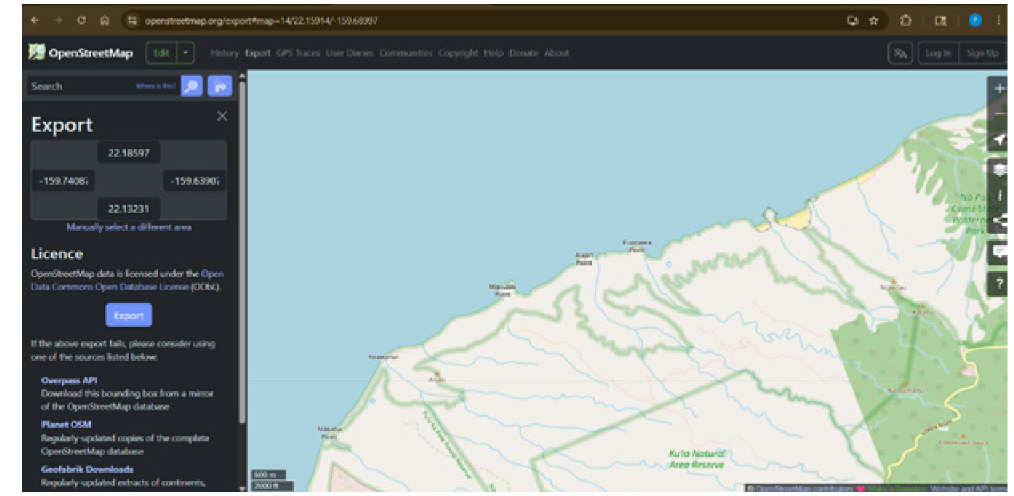
1. Gathering Site LiDAR information

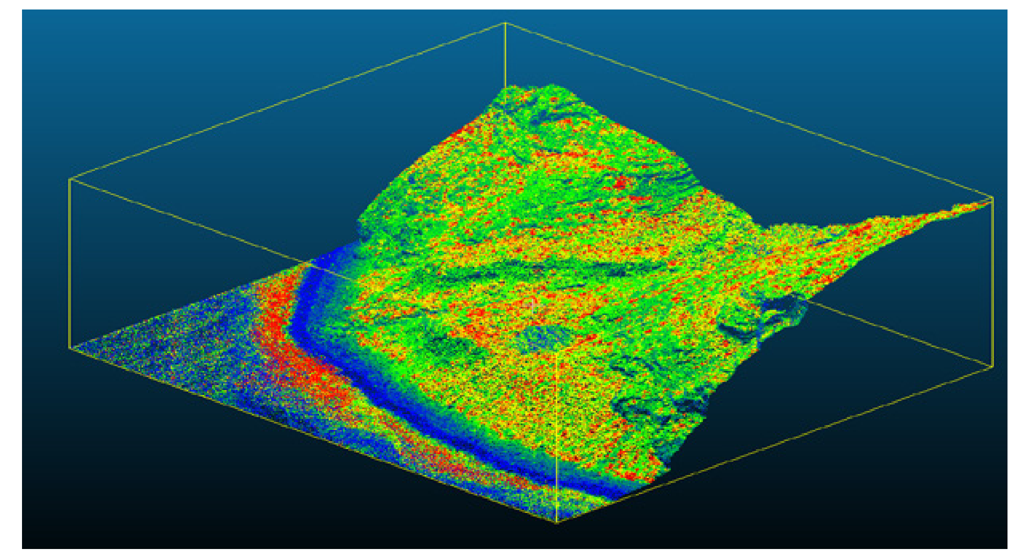
Depending on where your project location is there are many ways to locate LiDAR data from around the world. For this project I was looking at site context in rural Hawaii, Open Street Maps was able to provide the necessary data.



Open Street Maps: Source data

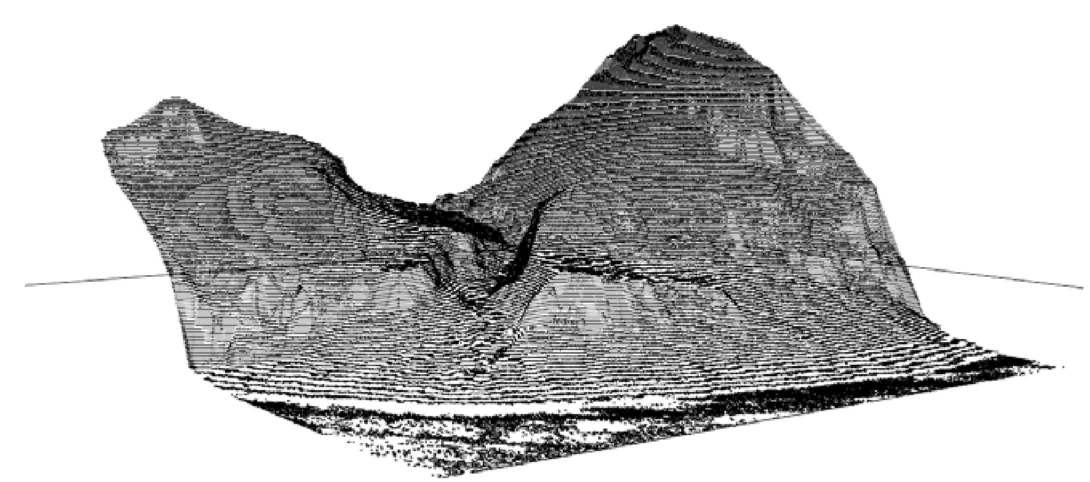
2. Refining Information in CloudCompare

Once LiDAR data is sourced it's time to bring it into Cloud Compare, here it is easy to refine the point cloud and seperate the data that you desire.


Cloud Compare: LiDAR refinement

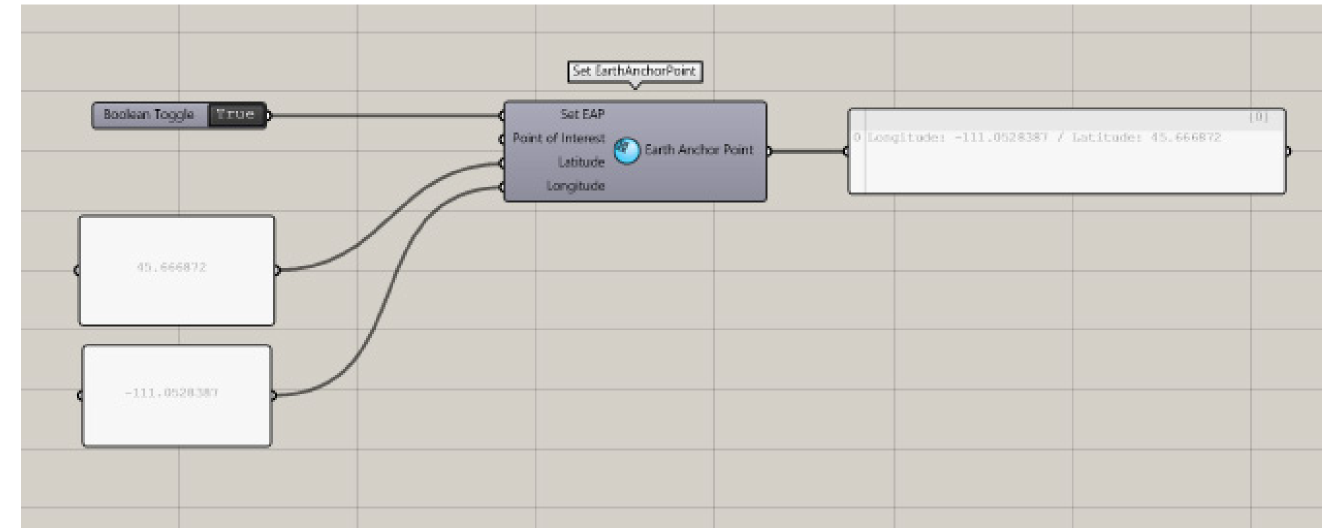
3. Transfer data to Rhino

Import refined point cloud into Rhino. Use countour command to crate line topography.


Rhino Topography

4. Set Earth Anchor Point

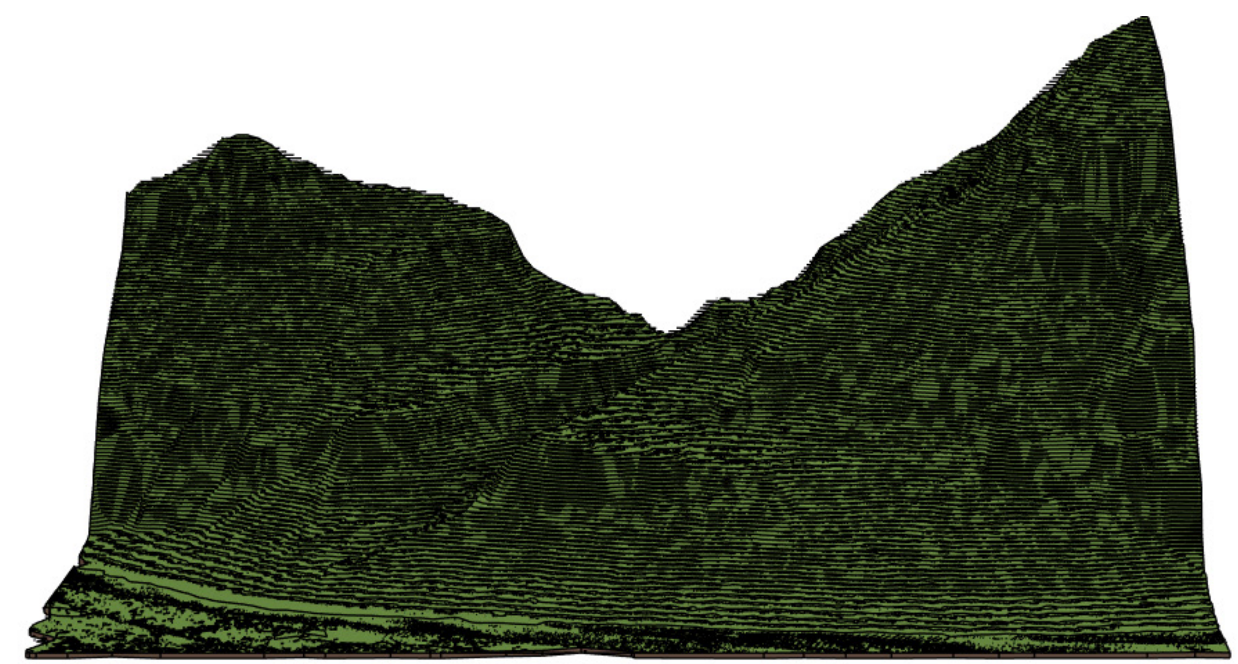
Using Grasshopper set an EAP (Earth Anchor Point) to ground for additional site information to remain in the correct location.



Grasshopper: Set Earth Anchor Point

5. Create Topo Solid in Revit

If needed bring file into Revit to create topo solid.

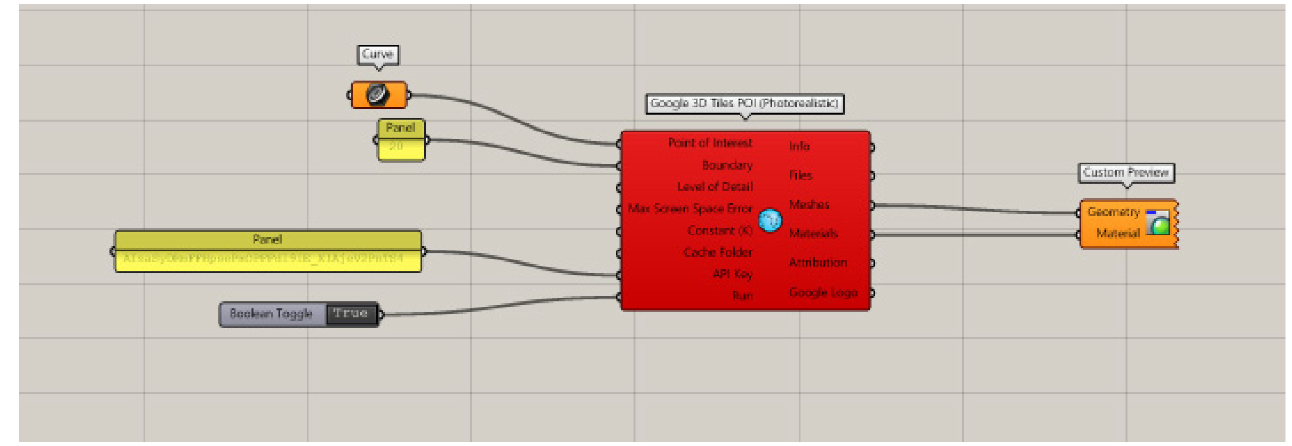

Revit: Create Topo Solid

7. Heron: Site Context

For this rural site there is no additional context to be gained.

7. Google Tiles

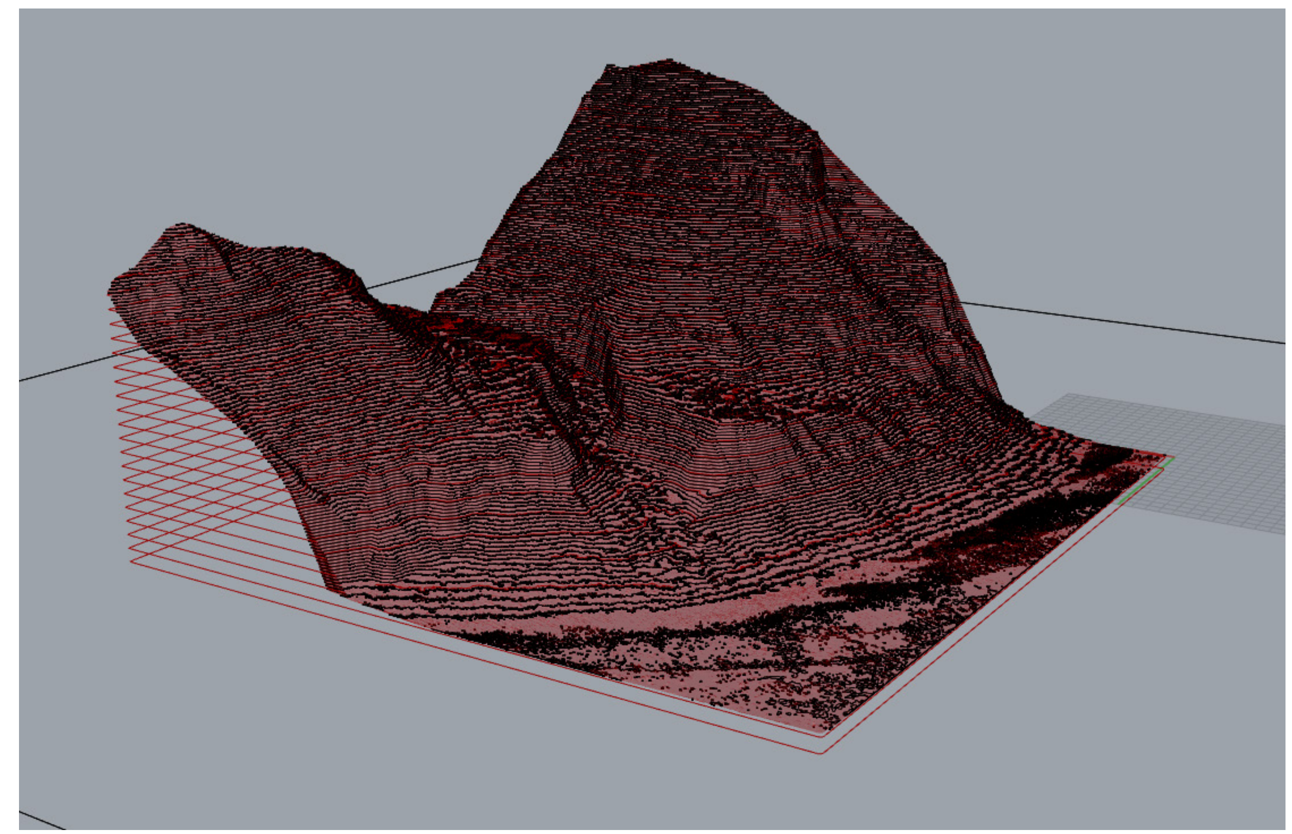
For this location there was no information for Google Tiles, though for many sites the context gained from GT could be very useful.

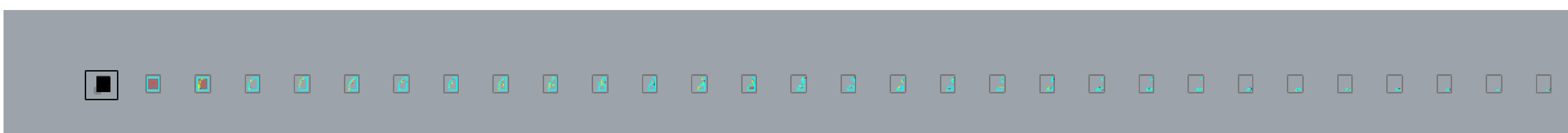


Google Tiles

8. Laser Cut: Physical Model

For the purpose of creating a physical model Grasshopper Open Nest command can be utilized to split topography and layout for lazer cutting.


Topo Base


Laser Cut Prep